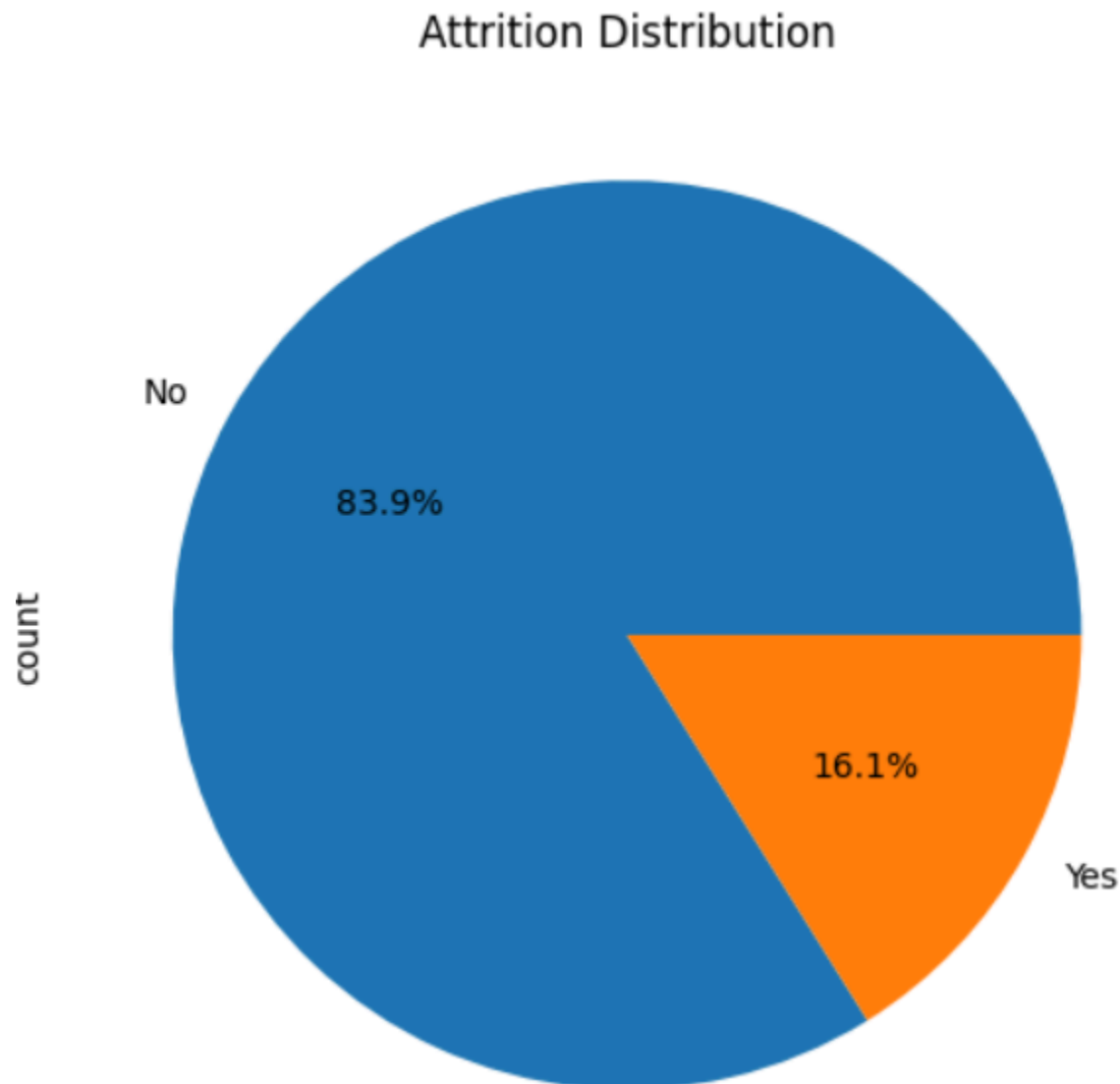


Initial Business Insights

A. Five Workforce Observations

1. The company has a stable workforce.

Approximately 83.9% of employees remain with the company, while 16.1% have left. This indicates that most employees stay, although attrition is still a concern.



2. Research & Development is the largest department.

The Research & Development department has the highest number of employees, followed by the Sales department. Human Resources has the smallest workforce.

3. Most employees do not work overtime.

The majority of employees reported not working overtime, suggesting that overtime is limited to a smaller group of the workforce.

4. The workforce is relatively young and experienced.

The average employee is about 37 years old, with an average of 11 years of total working experience and approximately 7 years at the company.

5. Most employees travel only occasionally.

The majority of employees fall under the Travel_Rarely category, indicating that frequent business travel is uncommon.

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	HourlyRate	JobInvolvement	JobLevel
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000
mean	36.923810	802.485714	9.192517	2.912925	1.0	1024.865306	2.721769	65.891156	2.729932	2.063946
std	9.135373	403.509100	8.106864	1.024165	0.0	602.024335	1.093082	20.329428	0.711561	1.106940
min	18.000000	102.000000	1.000000	1.000000	1.0	1.000000	1.000000	30.000000	1.000000	1.000000
25%	30.000000	465.000000	2.000000	2.000000	1.0	491.250000	2.000000	48.000000	2.000000	1.000000
50%	36.000000	802.000000	7.000000	3.000000	1.0	1020.500000	3.000000	66.000000	3.000000	2.000000
75%	43.000000	1157.000000	14.000000	4.000000	1.0	1555.750000	4.000000	83.750000	3.000000	3.000000
max	60.000000	1499.000000	29.000000	5.000000	1.0	2068.000000	4.000000	100.000000	4.000000	5.000000

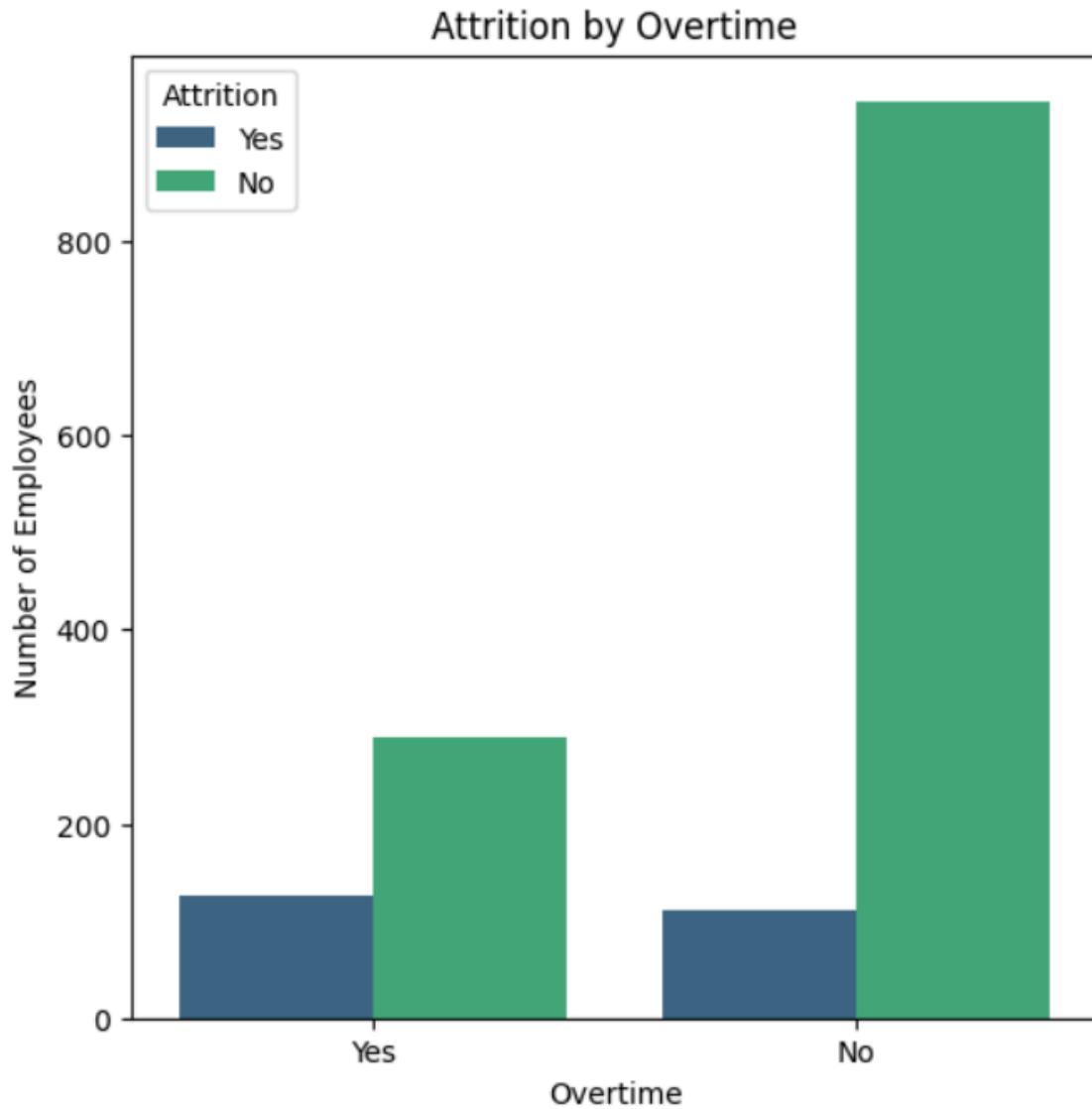
rows × 26 columns

	Attrition	BusinessTravel	Department	EducationField	Gender	JobRole	MaritalStatus	Over18	OverTime
count	1470	1470	1470	1470	1470	1470	1470	1470	1470
unique	2	3	3	6	2	9	3	1	2
top	No	Travel_Rarely	Research & Development	Life Sciences	Male	Sales Executive	Married	Y	No
freq	1233	1043	961	606	882	326	673	1470	1054

B. Five Possible Factors Contributing to Attrition

1. Overtime may increase employee attrition.

Employees who work overtime appear more likely to leave the company than employees who do not work overtime. This suggests that long working hours may contribute to stress and lower job satisfaction.

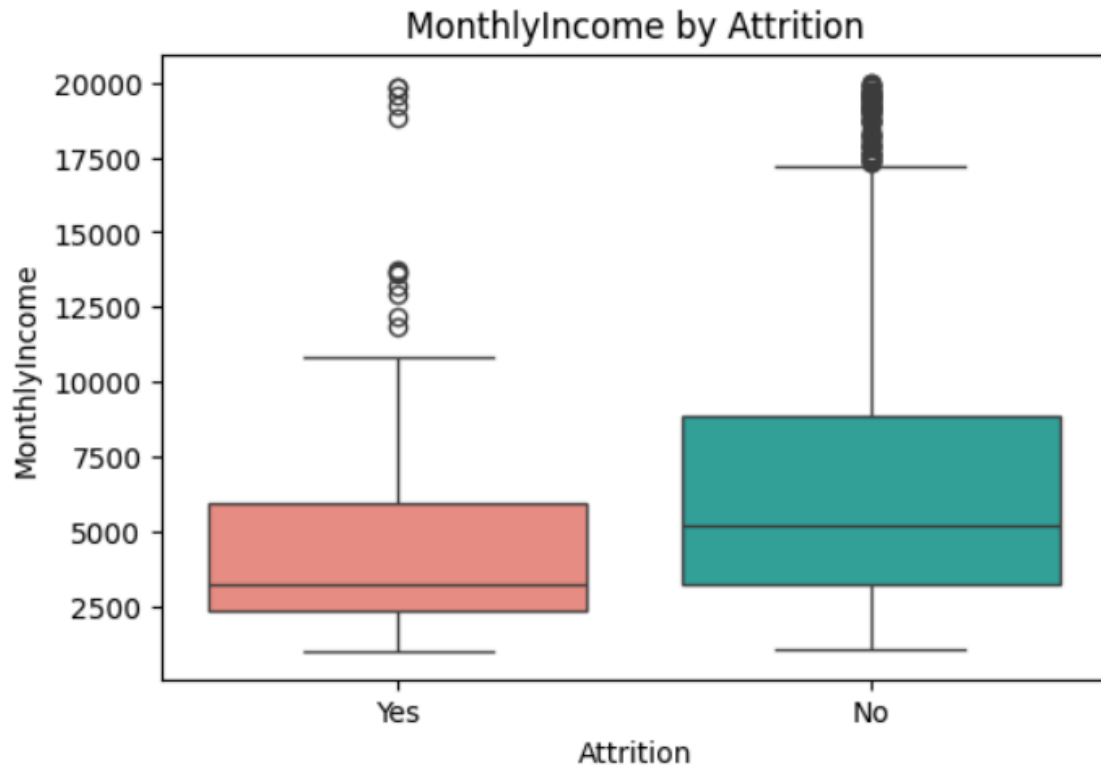


2. Younger employees are more likely to leave.

The boxplots show that employees who left the company are generally younger than those who stayed. Younger employees may be more likely to seek career growth or better opportunities elsewhere.

3. Lower monthly income is associated with higher attrition.

Employees who left tend to earn lower monthly salaries than employees who remained. This suggests that compensation may influence an employee's decision to stay.



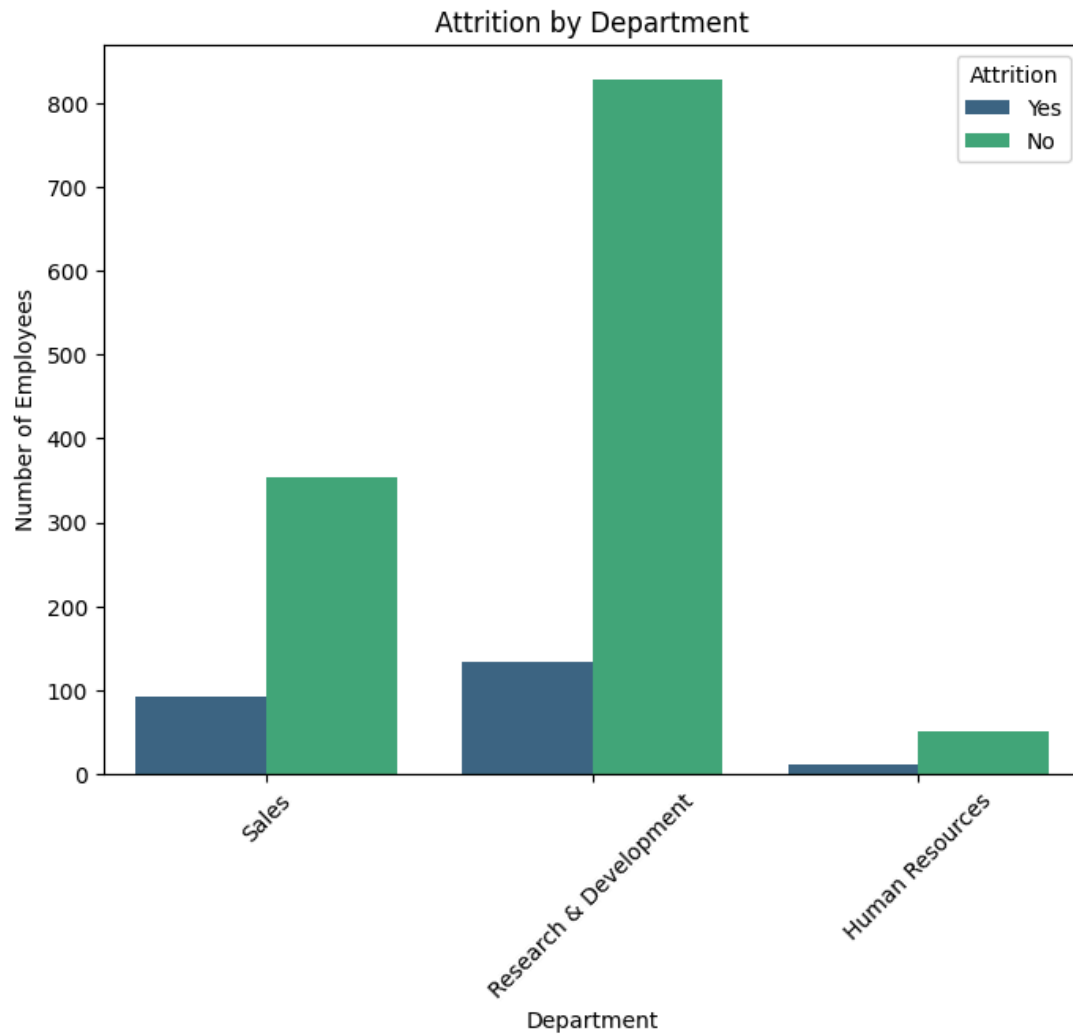
4. Employees with shorter tenure are more likely to leave.

Employees with fewer YearsAtCompany, YearsInCurrentRole, and TotalWorkingYears show higher attrition. This indicates that newer and less experienced employees are more likely to resign.

5. Certain departments and job roles experience higher turnover.

The Research & Development and Sales departments, particularly roles such as Laboratory Technician, Sales Executive, and Research Scientist, show relatively higher numbers of employees leaving. These positions may require additional attention to improve employee retention.

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All the relevant charts are in the Jupyter notebook

Conclusion

The exploratory analysis suggests that overtime, age, salary, work experience, length of service, and job role are the variables most closely associated with employee attrition. These factors should be investigated further and considered when developing future predictive machine learning models to identify employees who may be at risk of leaving.

Alia Al-Qadri